Subtitle Basic idea about DC to AC converter
Institute Dr B C Roy Engineering College, Durgapur
Subject Power Electronics EE 603

Inverter Lecture 05

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Hello friends

Lecture 05

1. Brief recap of last lecture 04
2. A brief review of PWM
3. Multiple pulse width modulation Inverter
4. Signification
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A brief review of PWM

- PWM:
  Pulse width modulation (PWM), or pulse-duration modulation (PDM), is a method of reducing the average power delivered by an electrical signal, by effectively chopping it up into discrete parts. The average value of voltage (and current) fed to the load is controlled by turning the switch between supply and load on and off at a fast rate. The longer the switch is on compared to the off periods, the higher the total power supplied to the load. [https://en.wikipedia.org/]

NB:

1. https://www.youtube.com/watch?v=YmPziPfaByw
2. https://www.multisim.com/content/DYJKviXg8EJwKGPvzPRXv5/pwmbyic555_km/open/

PWM inverter

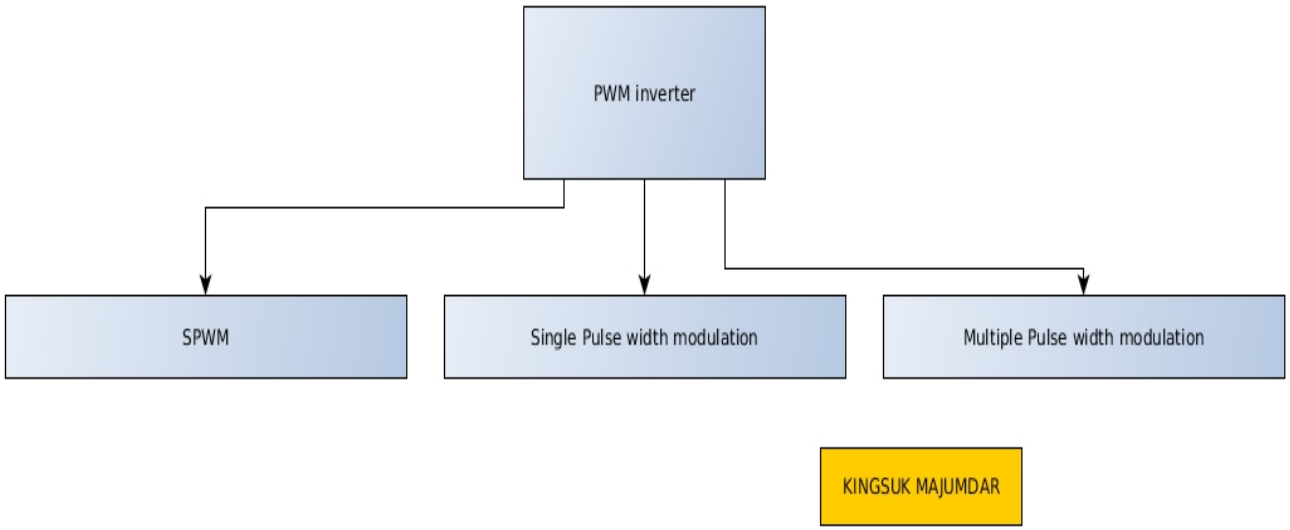


Figure: PWM inverter type

Multiple pulse width modulation Inverter

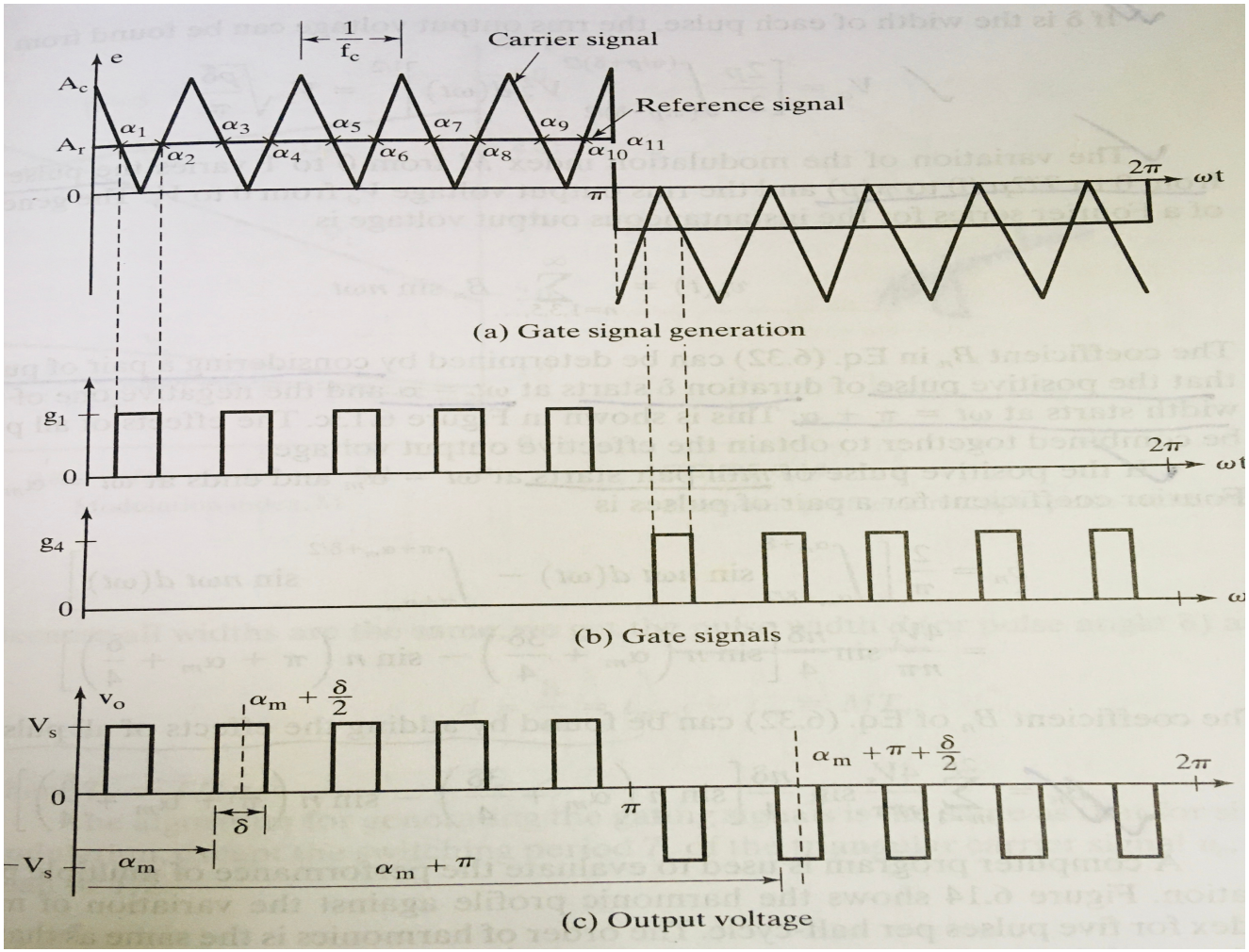


Figure: Multiple pulse width modulation Inverter

NB:picture taken from book [1]

AM index and FM index

- AM index (Amplitude modulation index)

$$M_a = \frac{A_{ref}}{A_{car}} \tag{1}$$

- FM index (Frequency modulation index)

$$M_f = \frac{f_{car}}{f_{out}} \tag{2}$$

- N,no of pulse per half cycle
  another name: Uniform pulse width modulation Inverter

$$N = \frac{f_{car}}{2 * f_{out}} = \frac{M_f}{2} \tag{3}$$

- The instantaneous output voltage

$$v_o = V_s(g_1 - g_4) \tag{4}$$

cont...

relation between pulse width and space can be realised

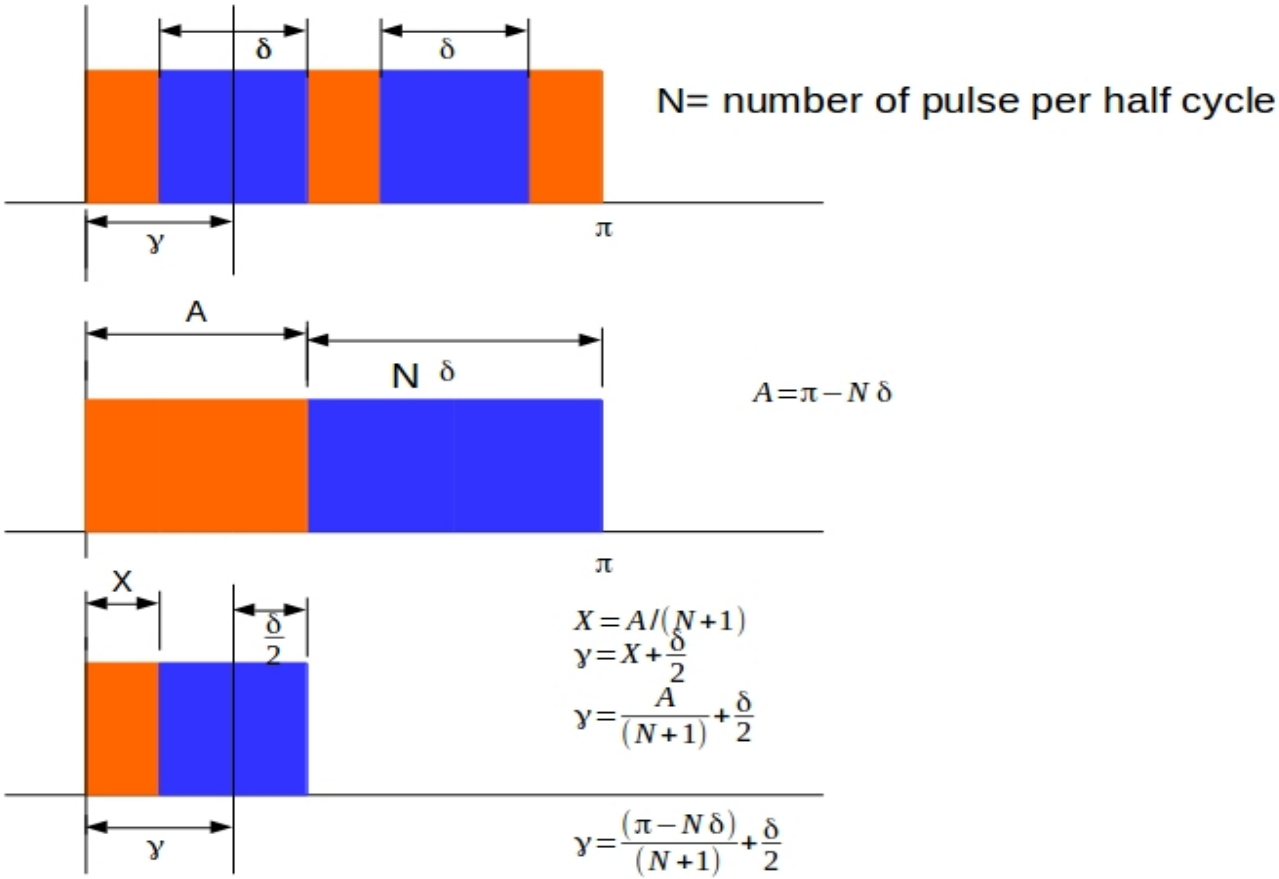


Figure: realisation of pulse and space gaps for mathematics

Circuit for Multiple pulse width modulation Inverter

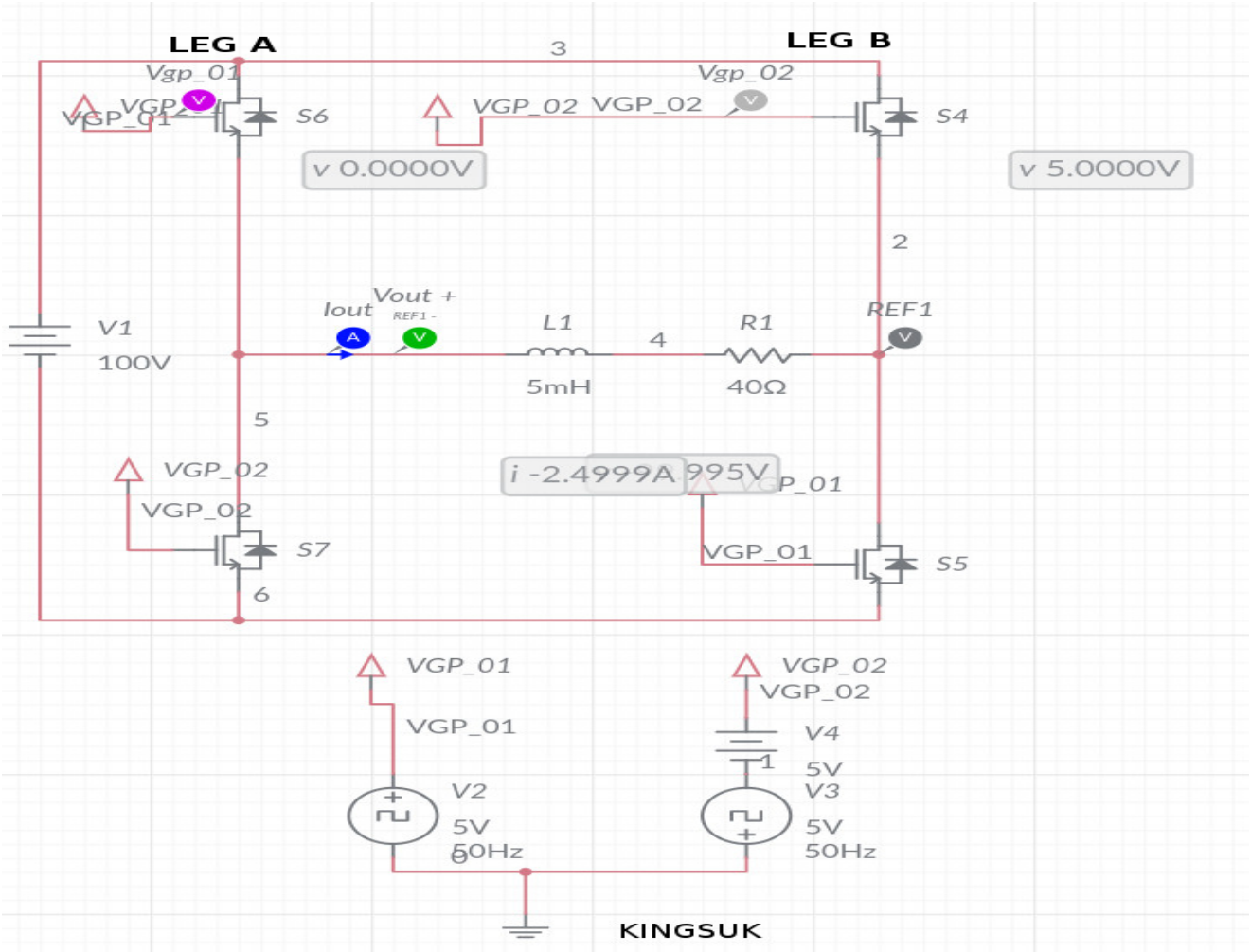


Figure: Circuit for Multiple PWM inverter

Multiple PWM mathematics

As the output waveform has quasi-square symmetry, hence, DC and cosine terms in the Fourier's series will be zero. take $\delta = 2 * d$ pulse width of the output voltage waveform. N= number of pulse per half cycle.

$$b_n = \frac{2}{\pi}\left[\int_{\gamma+\frac{\delta}{2}}^{\gamma+\delta} \sin(n\omega t)d(\omega t) - \int_{\pi+\gamma}^{\pi+\gamma+\frac{\delta}{2}} \sin(n\omega t)d(\omega t)\right] \tag{5}$$

$$b_n = \frac{4V_s}{n\pi}\sin\frac{n\delta}{4}\left[\sin n\left(\gamma+\frac{3\delta}{4}\right) - \sin n\left(\pi+\gamma+\frac{\delta}{4}\right)\right] \tag{6}$$

cont...

for all pulses B_n

$$B_n = \sum_{m=1}^{2N} b_n \tag{7}$$

Hence, instantaneous output voltage in Fourier's form

$$v_0(t) = \sum_{n=1,3,5...}^{\text{inf}} B_n \sin(n\omega t) \tag{8}$$

The rms output voltage

$$V_{rms} = Vs\sqrt{\frac{N\delta}{\pi}} \tag{9}$$

Signification:

- from eq: 8 instantaneous voltage can be written simply as follows

$$v_o = \sum_{n=1,3,5\ldots}^{\inf} \frac{4NV_s}{n\pi} \sin(n\gamma) \sin(n\frac{\delta}{2}) \sin(n\omega t) \tag{10}$$

- Amplitude of nth order harmonics

$$V_{on} = \frac{4NV_s}{n\pi} \sin(n\gamma) \sin(n\frac{\delta}{2}) \tag{11}$$

  Hence from eq 11 $V_o n$ depends on γ and δ, and N

- To mitigate a particular harmonic say n^{th} harmonic take $\delta = \frac{2\pi}{n}$ and $n\gamma = \pi$

cont..

The magnitude of 3rd harmonic and N= 2 (say) and $\delta = 36^0(taken)$

$$\gamma = \frac{180 - 2 * 36}{2 + 1} + \frac{\delta}{2}$$

$$\gamma = 54^0$$

from eq 11

$$V_{o1m} = 0.637 V_s \tag{12}$$

References


M. H. Rashid, *Power electronics handbook: devices, circuits and applications*. Elsevier, 2010.


N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics. Converters, Applications and Design*, 3rd ed. John Wiley and Sons, Inc, 2003.


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